

Assignment 3: Data Exploration

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change 'Student Name' on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. [NEW] Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to Canvas.
8. Initial here to acknowledge that you did not use AI in completing this assignment, except where expressly allowed: LZ

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks in your code chunks.

TIP: If your code fails to knit, check: * That no `install.packages()` or `View()` commands exist in your code. * That you are not displaying the entire contents of a large dataframe in your code.

Set up your R session

1. Load necessary packages (tidyverse, here), check your current working directory and import two datasets: the ECOTOX neonicotinoid dataset (`ECOTOX_Neonicotinoids_Insects_raw.csv`) and the Niwot Ridge NEON dataset for litter and woody debris (`NEON_NIWO_Litter_massdata_2018-08_raw.csv`). Name these datasets 'Neonics' and 'Litter', respectively.

Be sure to: * Use the `here()` package in specifying the paths to your datasets * Include the appropriate subcommand to read in character based columns as factors

```
# Load two necessary packages.
library(here)
library(tidyverse)
```

```
# Introduce two dataframes.
getwd()
```

```
## [1] "C:/Users/24169/Desktop/e872/872_spring26"
```

```
Neonics <- read.csv(file = here('./Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv'), stringsAsFactors = 
Litter <- read.csv(file = here('./Data/Raw/NEON_NIW0_Litter_massdata_2018-08_raw.csv'), stringsAsFactor
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information. (AI is allowed here, but put answers in your own words.)

Answer: Neonicotinoids can kill insects indiscriminately, and honeybees have encountered considerable threats. In addition, neonicotinoids can penetrate into plant cells and remain in water, which means that their presence in the natural environment is extensive. The chronic risk of neonicotinoids to humans is also possible.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information. (AI is allowed here, but put answers in your own words.)

Answer: Forest litters are associated with the wellness of surrounding ecosystems. They may reflect biodiversity, climate change, or decomposition patterns.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1. The litter sampling is targeted to take place in numerous 40mx40m plots. 2. Ground traps are sampled once per year. 3. No single site should ever exceed 3440 data product instances in a given calendar year.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
# Determine the dimension of 'Neonics'. There are 4623 rows and 30 columns.
str(Neonics)
```

```
## 'data.frame': 4623 obs. of 30 variables:
## $ CAS.Number : int 58842209 58842209 58842209 58842209 58842209 58842209 58842209 58842209
## $ Chemical.Name : Factor w/ 9 levels "(1E)-N-[(6-Chloro-3-pyridinyl)methyl]-N'-cy
## $ Chemical.Grade : Factor w/ 9 levels "Analytical grade",...: 9 9 9 9 9 9 9 9 9
## $ Chemical.Analysis.Method : Factor w/ 5 levels "Measured","Not coded",...: 4 4 4 4 4 4 4 4 4
## $ Chemical.Purity : Factor w/ 80 levels ">=98",">=99.0",...: 69 69 50 50 50 50 50 50
## $ Species.Scientific.Name : Factor w/ 398 levels "Acalolepta vastator",...: 69 69 248 248 248
## $ Species.Common.Name : Factor w/ 303 levels "Alfalfa Leafcutter Bee",...: 74 74 142 142
## $ Species.Group : Factor w/ 4 levels "Insects/Spiders",...: 1 1 1 1 1 1 1 1 1
## $ Organism.Lifestage : Factor w/ 20 levels "Adult","Cocoon",...: 1 1 19 19 19 1 19 11
## $ Organism.Age : Factor w/ 39 levels "~10","~24","~7",...: 39 39 39 39 39 36 39 36
## $ Organism.Age.Units : Factor w/ 11 levels "Day(s)","Days post-emergence",...: 9 9 4 4 4
## $ Exposure.Type : Factor w/ 24 levels "Choice","Dermal",...: 23 23 11 11 11 11 11
## $ Media.Type : Factor w/ 10 levels "Agar","Artificial soil",...: 7 7 3 3 3 3 3
## $ Test.Location : Factor w/ 4 levels "Field artificial",...: 4 4 4 4 4 4 4 4
## $ Number.of.Doses : Factor w/ 30 levels "' 4-5',' 4-7',...: 30 30 18 18 18 18 18
## $ Conc.1.Type..Author. : Factor w/ 3 levels "Active ingredient",...: 1 1 1 1 1 1 1 1
## $ Conc.1..Author. : Factor w/ 1006 levels "~10","~30/","~40/",...: 639 510 813 622 4
## $ Conc.1.Units..Author. : Factor w/ 148 levels "%","% v/v","% w/v",...: 132 132 91 91 91 9
## $ Effect : Factor w/ 19 levels "Accumulation",...: 16 16 16 16 16 16 16 16
## $ Effect.Measurement : Factor w/ 155 levels "Abundance","Accuracy of learned task, per
## $ Endpoint : Factor w/ 28 levels "EC10","EC50",...: 15 15 8 8 8 8 8 8 8
## $ Response.Site : Factor w/ 19 levels "Abdomen","Brain",...: 14 14 14 14 14 14 14
## $ Observed.Duration..Days. : Factor w/ 361 levels "~.1458","~10",...: 145 145 145 145 145 145
## $ Observed.Duration.Units..Days. : Factor w/ 17 levels "Day(s)","Day(s) post-emergence",...: 1 1 1
## $ Author : Factor w/ 433 levels "Abbott,V.A., J.L. Nadeau, H.A. Higo, and
## $ Reference.Number : int 107388 107388 103312 103312 103312 103312 103312 103312
## $ Title : Factor w/ 458 levels "A Common Pesticide Decreases Foraging Suc
## $ Source : Factor w/ 456 levels "Acta Hortic.1094:451-456",...: 295 295 296
## $ Publication.Year : int 1982 1982 1986 1986 1986 1986 1986 1986 1986
## $ Summary.of.Additional.Parameters: Factor w/ 943 levels "Purity: \xca NR - NR | Organism Age: \xca
```

6. Using the `summary` function on the 'Effect' column, determine the most common effects that are studied.
 [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
# List the distribution of 'Effect' column in an ascending sequence.
sort(summary(Neonics$Effect))
```

```
##      Hormone(s)      Histology      Physiology      Cell(s)
##           1           5           7           9
##      Biochemistry      Accumulation      Intoxication      Immunological
##           11           12           12           16
##      Morphology      Growth      Enzyme(s)      Genetics
##           22           38           62           82
##      Avoidance      Development      Reproduction      Feeding behavior
##          102          136          197          255
##      Behavior      Mortality      Population
##          360          1493          1803
```

Question: Which two effects stand out as the most studied? Can you guess why these effects might specifically be of interest? > Answer: Population and mortality are the most studied. They might be vital effects that an insect undergoes. They might be relatively easier to observe.

- Using the `summary` function, determine the six most commonly studied species in the dataset (common name).[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
# List the distribution of 'Species.Common.Name' column in a descending sequence.
sort(summary(Neonics$Species.Common.Name, maxsum = 4623), decreasing = TRUE)
```

##	Honey Bee	Parasitic Wasp
##	667	285
##	Buff Tailed Bumblebee	Carniolan Honey Bee
##	183	152
##	Bumble Bee	Italian Honeybee
##	140	113
##	Japanese Beetle	Asian Lady Beetle
##	94	76
##	Euonymus Scale	Wireworm
##	75	69
##	European Dark Bee	Minute Pirate Bug
##	66	62
##	Asian Citrus Psyllid	Parastic Wasp
##	60	58
##	Colorado Potato Beetle	Parasitoid Wasp
##	57	51
##	Erythrina Gall Wasp	Beetle Order
##	49	47
##	Snout Beetle Family, Weevil	Sevenspotted Lady Beetle
##	47	46
##	True Bug Order	Buff-tailed Bumblebee
##	45	39
##	Aphid Family	Cabbage Looper
##	38	38
##	Sweetpotato Whitefly	Braconid Wasp
##	37	33
##	Cotton Aphid	Predatory Mite
##	33	33
##	Ladybird Beetle Family	Parasitoid
##	30	30
##	Scarab Beetle	Spring Tiphia
##	29	29
##	Thrip Order	Ground Beetle Family
##	29	27
##	Rove Beetle Family	Tobacco Aphid
##	27	27
##	Chalcid Wasp	Convergent Lady Beetle
##	25	25
##	Stingless Bee	Spider/Mite Class
##	25	24
##	Tobacco Flea Beetle	Citrus Leafminer
##	24	23
##	Ladybird Beetle	Mason Bee

##	23	22
##	Mosquito	Argentine Ant
##	22	21
##	Beetle	Flatheaded Appletree Borer
##	21	20
##	Horned Oak Gall Wasp	Leaf Beetle Family
##	20	20
##	Potato Leafhopper	Tooth-necked Fungus Beetle
##	20	20
##	Codling Moth	Black-spotted Lady Beetle
##	19	18
##	Calico Scale	Fairyfly Parasitoid
##	18	18
##	Lady Beetle	Minute Parasitic Wasps
##	18	18
##	Mirid Bug	Mulberry Pyralid
##	18	18
##	Silkworm	Vedalia Beetle
##	18	18
##	Araneoid Spider Order	Bee Order
##	17	17
##	Egg Parasitoid	Insect Class
##	17	17
##	Moth And Butterfly Order	Oystershell Scale Parasitoid
##	17	17
##	Hemlock Woolly Adelgid Lady Beetle	Hemlock Woolly Adelgid
##	16	16
##	Mite	Onion Thrip
##	16	16
##	Western Flower Thrips	Corn Earworm
##	15	14
##	Green Peach Aphid	House Fly
##	14	14
##	Ox Beetle	Red Scale Parasite
##	14	14
##	Spined Soldier Bug	Armoured Scale Family
##	14	13
##	Diamondback Moth	Eulophid Wasp
##	13	13
##	Monarch Butterfly	Predatory Bug
##	13	13
##	Yellow Fever Mosquito	Braconid Parasitoid
##	13	12
##	Common Thrip	Eastern Subterranean Termite
##	12	12
##	Jassid	Mite Order
##	12	12
##	Pea Aphid	Pond Wolf Spider
##	12	12
##	Spotless Ladybird Beetle	Glasshouse Potato Wasp
##	11	10
##	Lacewing	Southern House Mosquito
##	10	10
##	Two Spotted Lady Beetle	Ant Family

##	10	9
##	Apple Maggot	Asiatic Honey Bee
##	9	9
##	Eulophid Parasitoid	Lacewing Family
##	9	9
##	Mealybug Destroyer	Alfalfa Leafcutter Bee
##	9	8
##	Bee	Bumblebee
##	8	8
##	Chilean Predatory Mite	Dwarf Honey Bee
##	8	8
##	Neotropical Stingless Bee	Parasitic Wasp Family
##	8	8
##	Spiralling Whitefly	Beetle Mite Family
##	8	7
##	Chinch Bug	Macedonian Honey Bee
##	7	7
##	Moth	Potato Tuberworm
##	7	7
##	Russian Wheat Aphid	Soldier Beetle
##	7	7
##	Southern One-Year Canegrub	Tarnished Plant Bug
##	7	7
##	Ambrosia Beetle	Aphid Wasp
##	6	6
##	Black Vine Weevil	Childers Canegrub
##	6	6
##	Coconut Leaf Beetle	Eleven-spotted Ladybird Beetle
##	6	6
##	Encyrtid Wasp	European Red Mite
##	6	6
##	Fall Armyworm	Fruit Fly
##	6	6
##	Hover Fly	Oblique Banded Leaf Roller
##	6	6
##	Obscure Mealybug	Oribatid Mite Suborder
##	6	6
##	Pistachio Psyllid	Redbay Ambrosia Beetle
##	6	6
##	Silverleaf Whitefly	Soybean Aphid
##	6	6
##	Subterranean Termite	Thrip
##	6	6
##	Two-Spotted Spider Mite	Apple Aphid
##	6	5
##	Brown Planthopper	Earwig
##	5	5
##	Green June Beetle	Hornfaced Bee
##	5	5
##	Long Horned Beetle Family	Plum Curculio
##	5	5
##	Rove Beetle	San Jose Scale
##	5	5
##	Scelionid Wasp	Speckled Cutworm Moth

##	5	5
##	Thrip Family	Ant
##	4	4
##	Cabbage Seedpod Weevil	Common Green Lacewing
##	4	4
##	Eucalyptus Gall Wasp	European Apple Sawfly
##	4	4
##	European Honey Bee	European Tarnished Plant Bug
##	4	4
##	Garden Symphytan	Linyphiid Spider
##	4	4
##	Onion Maggot	Oriental Beetle
##	4	4
##	Parsnip Seed Wasp	Pea And Bean Weevil
##	4	4
##	Pear Sucker	Red Imported Fire Ant
##	4	4
##	Striped Cucumber Beetle	Sugarcane Beetle
##	4	4
##	Wasp	Wolf Spider Family
##	4	4
##	Yellow-faced Bumblebee	Ambrosia Bark Beetle
##	4	3
##	Asian Ambrosia Beetle	Beetle Family
##	3	3
##	Birch Leafminer	Black Twig Borer
##	3	3
##	Braconid Parasitoid Wasp	California Red Scale
##	3	3
##	Crucifer Flea Beetle	Cutworm
##	3	3
##	Delphacid Planthopper	Egyptian Cotton Leafworm
##	3	3
##	Encyrtid Parasitoid	Fly/Mosquito/Midge Order
##	3	3
##	Formosan Subterranean Termite	Fruit-tree Pinhole Borer
##	3	3
##	Green Rice Leafhopper	Ground Beetle
##	3	3
##	Ichneumonid Wasp	Large-Jawed Orb Weaver Family
##	3	3
##	Leaf Cutting Ant	Mediterranean Fruit Fly
##	3	3
##	Minute Flour Bug	Mite Family
##	3	3
##	Moth Family	Negatoria Canegrub
##	3	3
##	Sap Beetle Family	Scale Insect Order
##	3	3
##	Scarab Beetle Family	Sheet-Web Weaver Family
##	3	3
##	Spider	Sugarcane Grub
##	3	3
##	Tenebrionid Beetle	Alfalfa Plant Bug

##	3	2
##	Alkali Bee	Aphid
##	2	2
##	Assassin Bug	Azalea Lace Bug
##	2	2
##	Banana Aphid	Brown Scale
##	2	2
##	Brown Stinkbug	Budworm
##	2	2
##	Cabbage Aphid	Cabbage White
##	2	2
##	Cardamom Thrip	Carrot Weevil
##	2	2
##	Celer Crab Spider	Centipede Class
##	2	2
##	Citricola Scale	Clouded Plant Bug
##	2	2
##	Coffee Bean Weevil	Cotton Fleahopper
##	2	2
##	Egyptian Alfalfa Weevil	Engraver Beetle
##	2	2
##	Fig Longicorn Beetle	Glassy-winged Sharpshooter
##	2	2
##	Hawthorn Lace Bug	Hister Beetle Family
##	2	2
##	Jumping Spider Family	Lined Click Beetle
##	2	2
##	Maple Spider Mite	Meshweaver Spider
##	2	2
##	Minute Pirate Bug Family	Predaceous Fly
##	2	2
##	Pygmy Mangold Beetle	Rose Sawfly
##	2	2
##	Serpentine Leafminer	Spider Mite Destroyer
##	2	2
##	Spotted Tentiform Leafminer	Stink Bug
##	2	2
##	Tawny Mole Cricket	Tick/Chigger/Mite Order
##	2	2
##	Turf Running-spider	Turnip Aphid
##	2	2
##	Western Bigeyed Bug	Western Damsel Bug
##	2	2
##	Western Plant Bug	White-backed Planthopper
##	2	2
##	White Apple Leafhopper Nymph	Whitemarked Fleahopper
##	2	2
##	Antlike Flower Beetle	Banded Soft-winged Flower Beetle
##	1	1
##	Banded Sunflower Moth	Bee Family
##	1	1
##	Beet Armyworm	Black Citrus Aphid
##	1	1
##	Blue Alfalfa Aphid	Cabbage Root Fly

##	1	1
##	Cactus Lady Beetle	Citrus Red Mite
##	1	1
##	Cottony Cushion Sale	Crapemyrtle Aphid
##	1	1
##	Damselbug Family	Ectoparasitoid Wasp
##	1	1
##	English Grain Aphid	Fairyfly
##	1	1
##	Flea Beetle	Gall Midge
##	1	1
##	Grasshopper/Cricket/Locust Order	Greenhouse Whitefly
##	1	1
##	Grey Sunflower Seed Weevil	Harvestman Spider Order
##	1	1
##	Hawthorn Leaf Miner	Longtailed Fruit Fly Parasite
##	1	1
##	Minute Lady Beetles	Painted Maple Aphid
##	1	1
##	Pepper Weevil	Pine False Webworm
##	1	1
##	Plant Bug	Pollen Beetle
##	1	1
##	Predacious Mite	Predator Bug
##	1	1
##	Pseudocentipede Class	Pteromalid Wasp Family
##	1	1
##	Red Sunflower Seed Weevil	Rice Leaf Folder Moth
##	1	1
##	Rose Grain Aphid	Scale Picnic Beetle
##	1	1
##	Shiny Spider Beetle	Southern Army Worm
##	1	1
##	Spirea Aphid	Spotted Sunflower Stem Weevil
##	1	1
##	Strawberry Blossom Weevil	Sunflower Midge
##	1	1
##	Sunflower Moth	Ten-spot Ladybird Beetle
##	1	1
##	Tobacco Thrip	Twicestabbed Lady Beetle
##	1	1
##	Wasp Family	Weevil
##	1	1
##	Yellow Mealworm Beetle	
##	1	

Question: What do these species have in common? Why might they be of interest over other insects? >
 Answer: Honey bee, parasitic wasp, buff tailed bumblebee, Carniolan honey bee, bumble bee, and Italian honeybee are the six species. They carry out significant economic benefits, so scholars might be interested in pursuing their research.

8. The `Conc.1..Author` column, which lists the concentration of the neonicitoid dose, should include numeric values. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
# Figure out the status of 'Conc.1..Author.' column.
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

Answer: It is factor. The values, containing integers, divisions, or 'NR', seem to be messy and inconsistent.

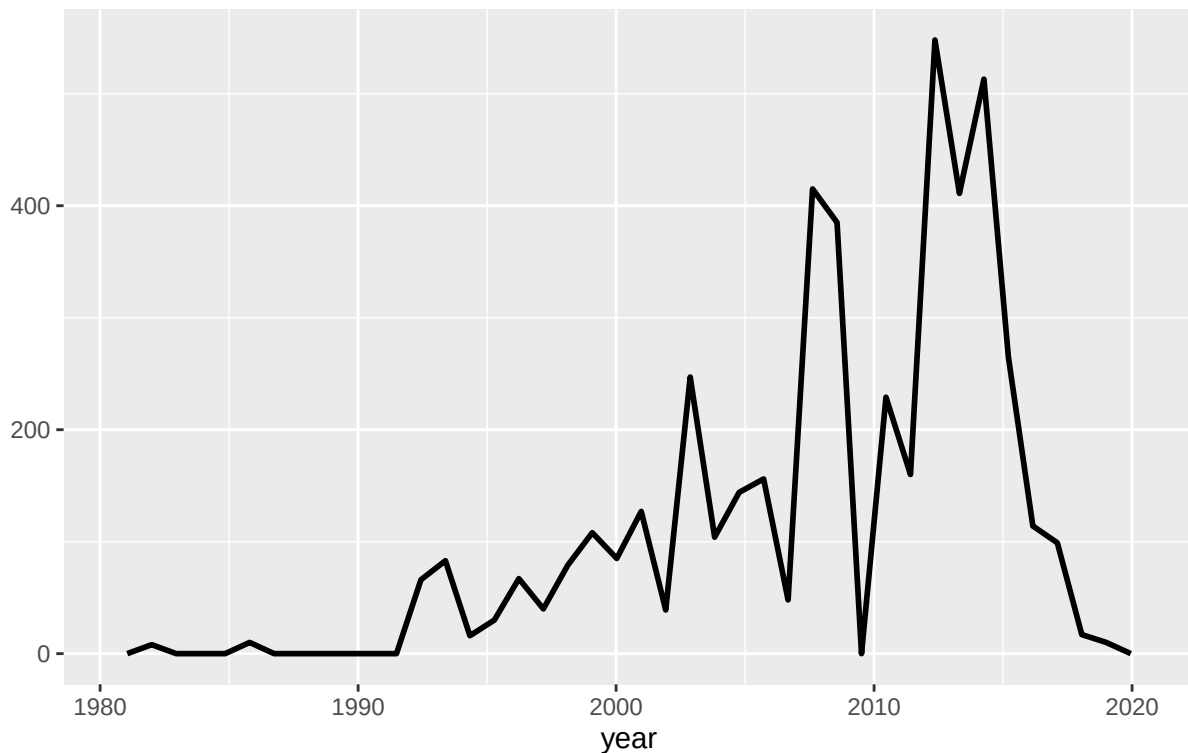
Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
# Derive a line plot regarding the frequency of 'Publication.Year' column.
p1 <- ggplot(data = Neonics, aes(x = Publication.Year)) + geom_freqpoly(bins = 40, size = 1, color = '#')
p1
```

Number Of Studies Each Year

Data From 4623 Observations



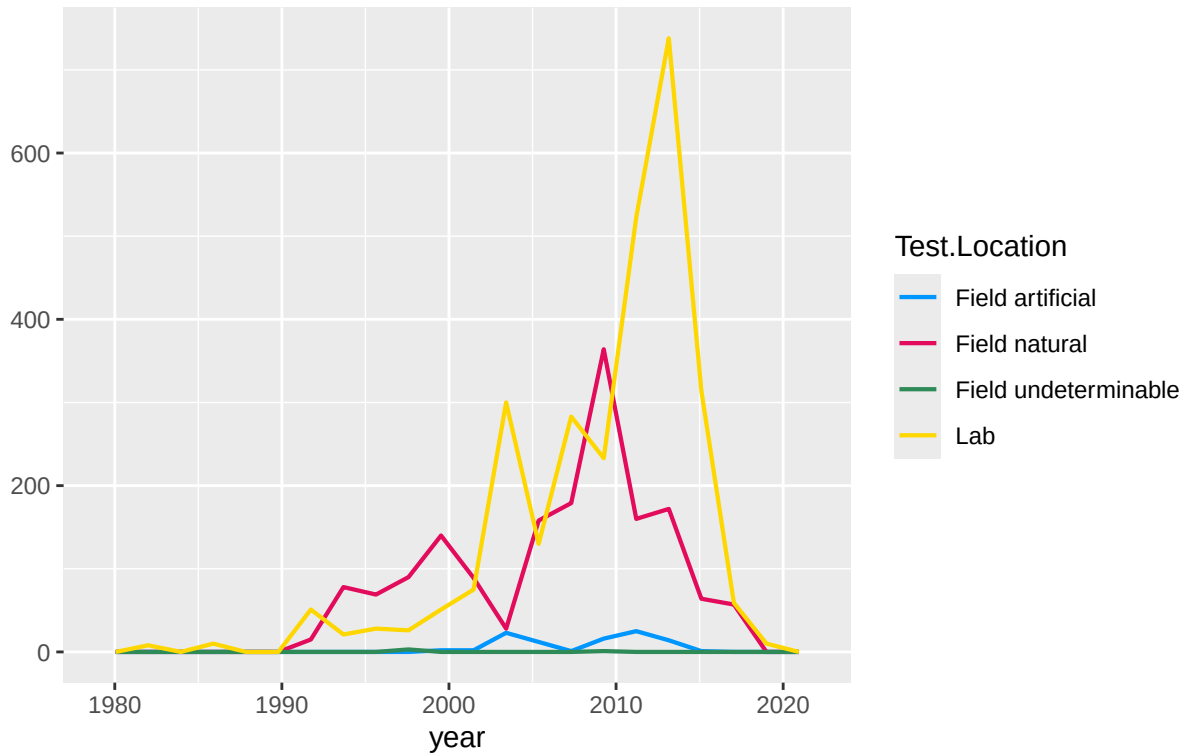
10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
# Determine the number of unique values in 'Test.Location' column.
unique(Neonics$Test.Location)
```

```
## [1] Lab                Field natural          Field artificial
## [4] Field undeterminable
## Levels: Field artificial Field natural Field undeterminable Lab
```

```
# Derive a differentiated line plot regarding the frequency of 'Publication.Year' column. A four-color
p2 <- ggplot(data = Neonics, aes(x = Publication.Year, color = Test.Location)) + geom_freqpoly(bins = 2)
p2
```

Number Of Studies Conducted At Different Locations Each Year
Data From 4623 Observations



Interpret this graph. What are the most common test locations, and do they differ over time? > Answer: Lab is the most common test location. Most lab tests took place after 2010, but most field natural tests took place before 2010.

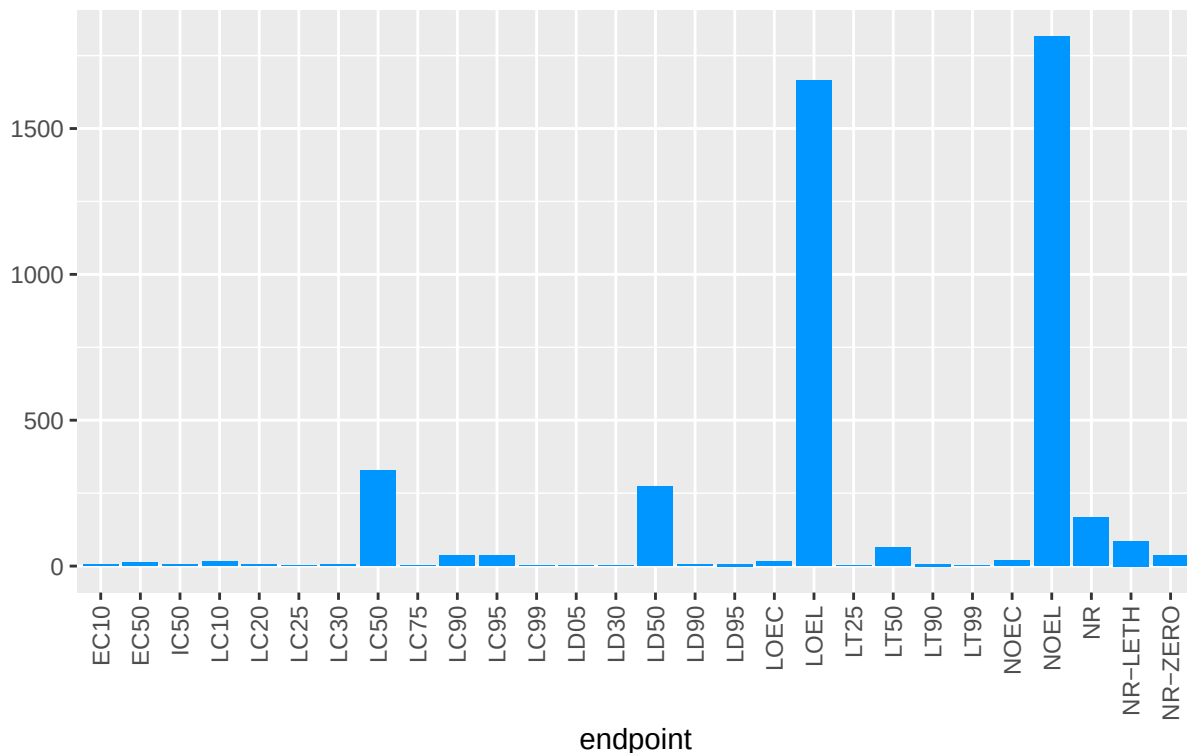
11. Create a bar graph of Endpoint counts.

[**TIP:** Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
# Derive a bar plot regarding the frequency of 'Endpoint' column.
p3 <- ggplot(data = Neonics, aes(x = Endpoint)) + geom_bar(fill = '#0096FF') + theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
p3
```

Number Of Studies Classified By Endpoint

Data From 4623 Observations



What are the two most common end points, and how are they defined? Consult the ECO-TOX_CodeAppendix (p.721) for more information. > Answer: NOEL and LOEL.

Explore your data (Litter)

- Determine the class of `collectDate`. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
# The first attempt shows that 'collectDate' column contains factor values.
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
# Change the format for 'collectDate' column. The second attempt shows that 'collectDate' column contains date values.
Litter$collectDate <- ymd(Litter$collectDate)
class(Litter$collectDate)
```

```
## [1] "Date"
```

```
# There are two dates for collection: 2018-08-02, 2018-08-30.
unique(Litter$collectDate)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different `plotIDs` sampled at Niwot Ridge.

```
# Discover the number of unique values in 'plotID' column.
```

```
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051  
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057  
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

```
summary(Litter$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061  
##      20      19      18      15      14      8      16      17  
## NIWO_062 NIWO_063 NIWO_064 NIWO_067  
##      14      14      16      17
```

How is the information obtained from `unique` different from that obtained from `summary`? > Answer: 'Summary' function shows all unique values and the frequency of each unique value. However, 'unique' function only shows all unique values.

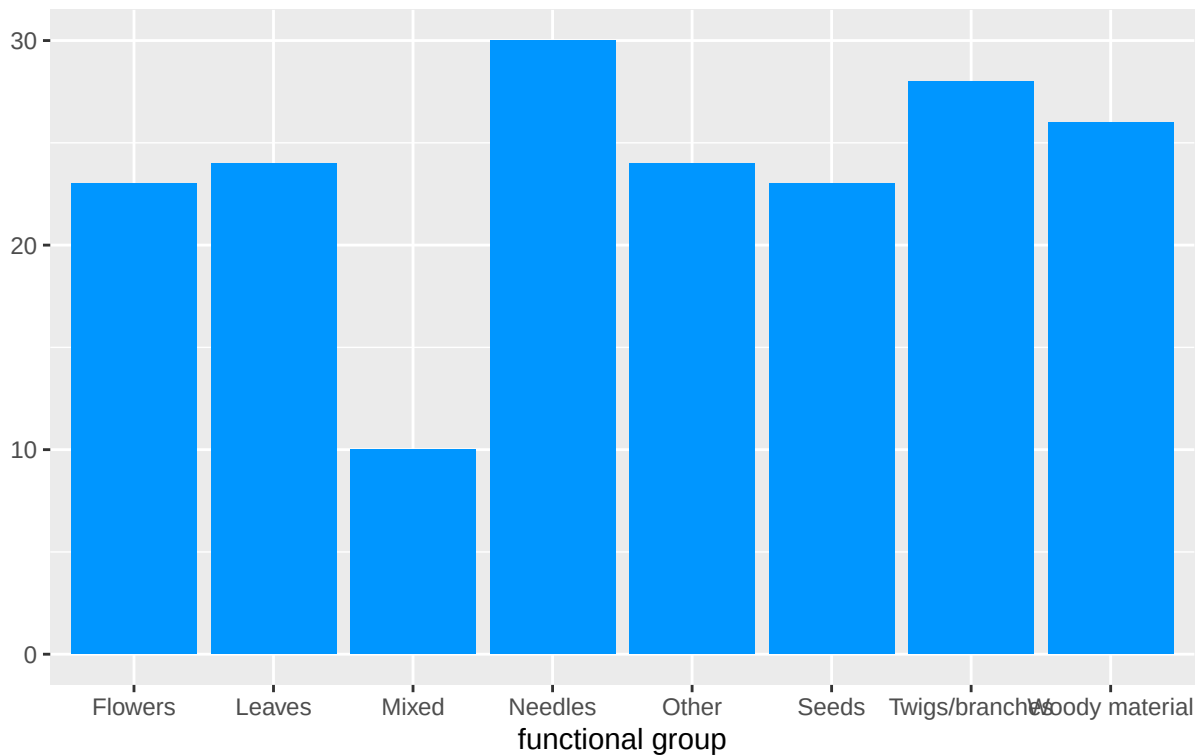
14. Create a bar graph of `functionalGroup` counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

```
# Derive a bar plot regarding the frequency of 'functionalGroup' column.
```

```
p4 <- ggplot(data = Litter, aes(x = functionalGroup)) + geom_bar(fill = '#0096FF') + labs(title = 'Number of  
p4
```

Number Of Records Classified By Functional Group

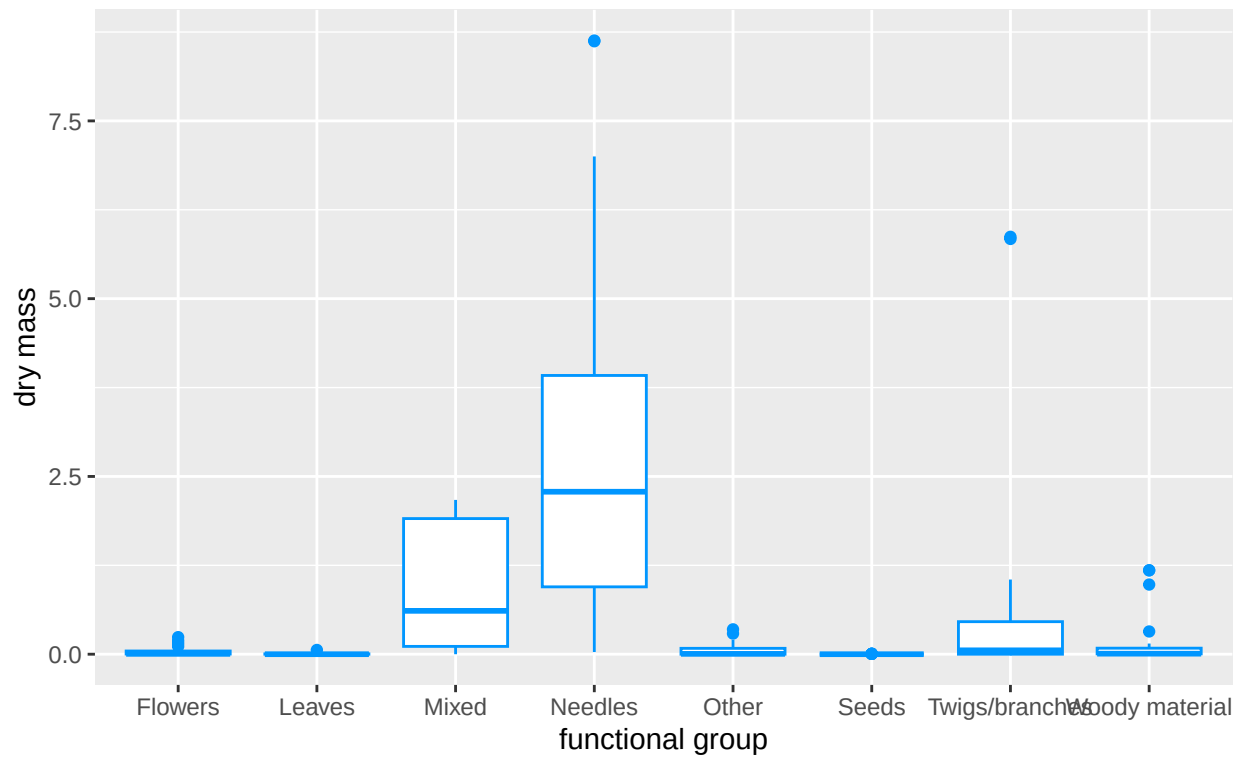
Data From 188 Observations



15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

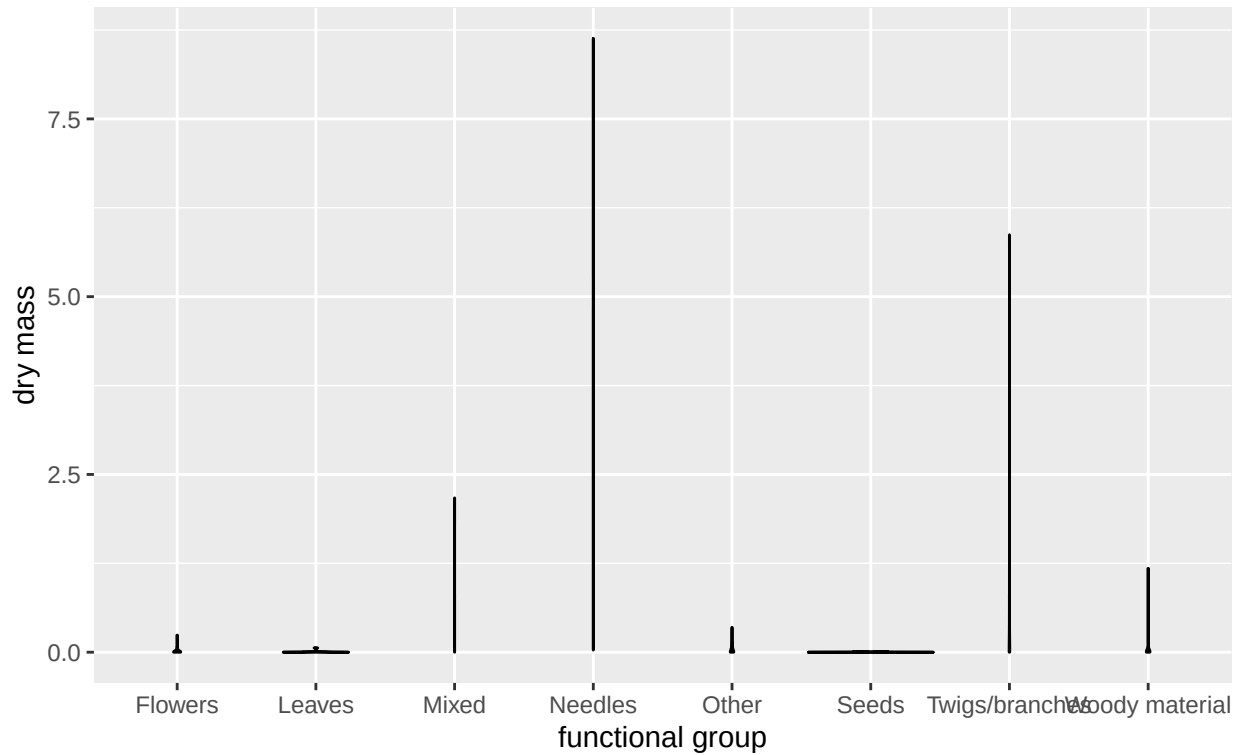
```
# Derive a box plot regarding the distribution of 'dryMass' column. The grouping factor is 'functionalGroup'  
p5 <- ggplot(data = Litter, aes(x = functionalGroup, y = dryMass)) + geom_boxplot(color = '#0096FF') +  
p5
```

Distribution Of Dry Mass Grouped By Functional Group
Data From 188 Observations



```
# Derive a violin plot regarding the distribution of 'dryMass' column. The grouping factor is 'functionalGroup'
p6 <- ggplot(data = Litter, aes(x = functionalGroup, y = dryMass)) + geom_violin(color = '#000000') + labs(y = 'dry mass')
p6
```

Distribution Of Dry Mass Grouped By Functional Group
Data From 188 Observations



Why is the boxplot a more effective visualization option than the violin plot in this case? > Answer: The sample size is too small for a violin plot to present clear information.

What type(s) of litter tend to have the highest biomass at these sites? > Answer: Needles tend to have the highest biomass.